JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RAMACHANDRA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

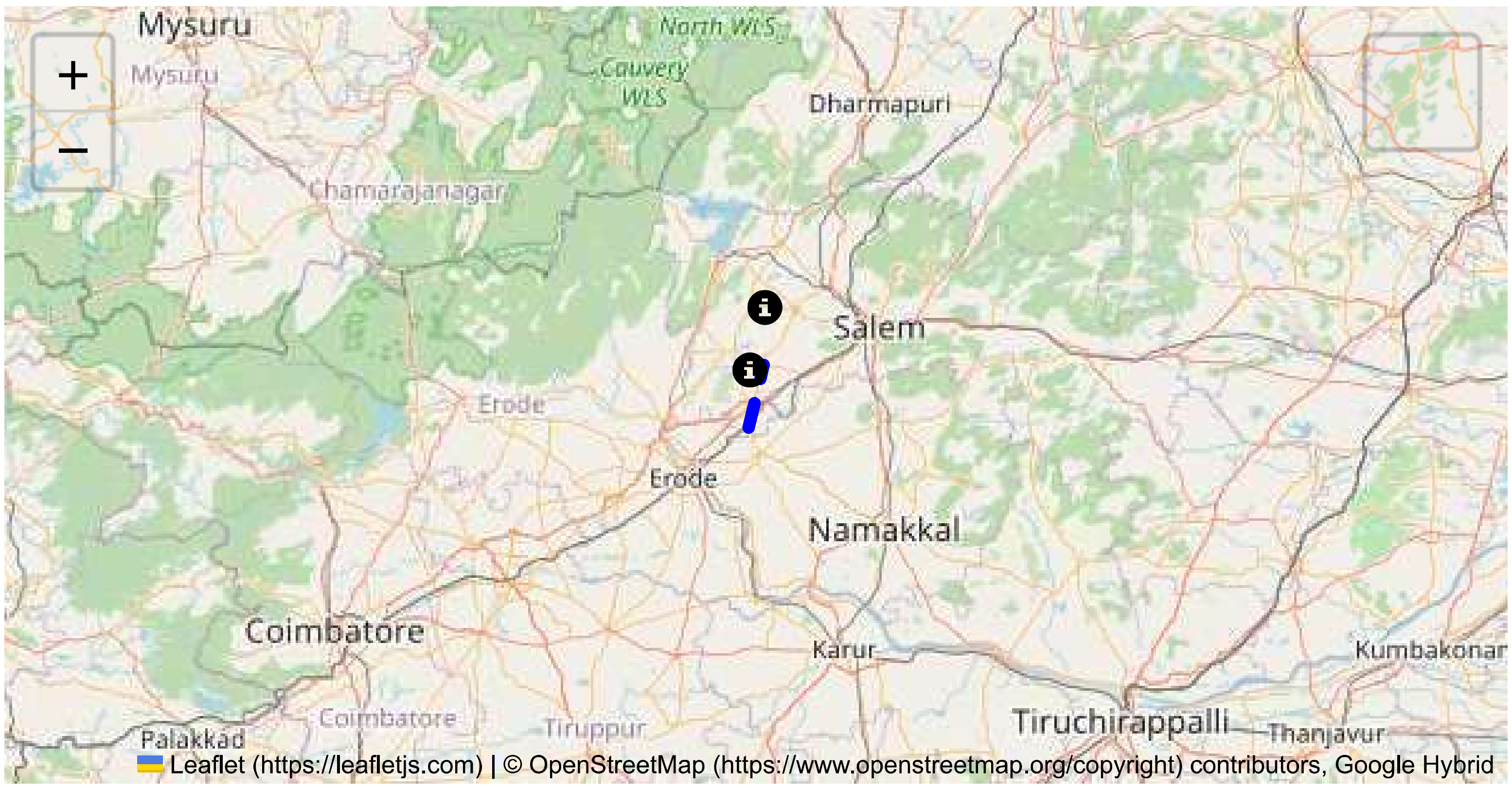
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

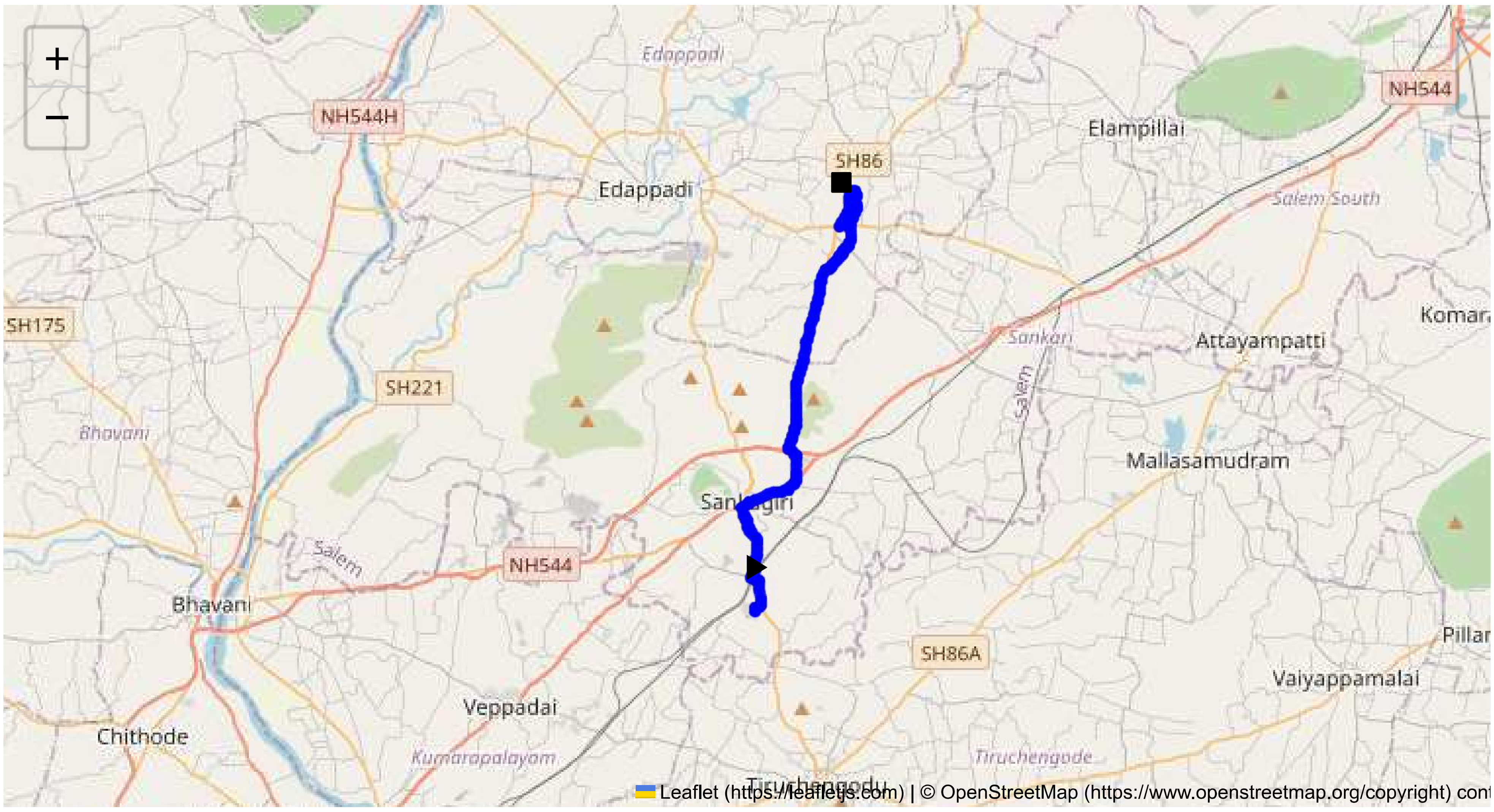
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 20.44 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.570888, 77.902984)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to HWC3+86V, Erumaipatti spans approximately 20.44 kilometers and typically takes about 56 minutes for heavy vehicles carrying hazardous materials. Starting from Sangagiri, drivers will primarily use local roads and intersect with SH 86 en route to Erumaipatti.

Typical Weather Conditions and Potential Hazards

Tamil Nadu experiences a tropical climate, with significant heat in summer months (March to June), monsoonal rains from October to December, and relatively mild winters (November to February). Heavy monsoonal rains can lead to reduced visibility, slick road surfaces, and potential for localized flooding, particularly in poorly drained areas.

Traffic Patterns

Traffic congestion is most likely to occur during traditional peak hours: 8:00 AM - 10:00 AM and 5:00 PM - 7:00 PM. Areas near intersections with State Highways and within towns such as Sangagiri typically experience higher congestion levels. Agricultural vehicles and local commutes can add to the traffic density during harvest periods.

Road Quality and Infrastructure

The road quality varies along the route. Within towns, roads may be narrower with frequent stops due to pedestrian crossings and local market activity, leading to increased potential for delays and incidents. Rural sections may lack roadside amenities, and occasional potholes or uneven surfaces can pose risks.

Alternative Routes for Emergencies

In an emergency, alternative routes might involve detours through smaller village roads or backtracking to larger highways for rerouting. However, these routes can be less reliable and potentially add significant time depending on road conditions.

Local Regulations for Hazardous Material Transport

The transport of hazardous materials is regulated under the Motor Vehicles Act and relevant safety protocols. Specific permissions should be gained prior to transport, including necessary permits, and compliant vehicles must meet safety standards. Restrictions may apply in densely populated areas or near schools.

Historical Incidents

Historically, incidents involving heavy vehicles on this route have involved collisions due to poor visibility and road conditions, particularly during monsoonal rains. While rare, there have been a few reported cases of hazardous material leaks due to accidents.

Environmental Considerations and Sensitive Areas

Key environmental considerations include protection of agricultural lands and minimizing emissions in populated areas. The route does not pass through designated protected environmental zones, but care should be taken to minimize pollution and disturbance.

Communication Coverage

Cellular coverage is generally reliable in urban and semi-urban areas; however, certain rural stretches may experience signal drop-outs. It is advisable to plan communication needs accordingly and consider emergency communication devices.

Emergency Response Times

Emergency response times can vary, with rural segments potentially experiencing longer wait times due to distance from major towns. In-town response times are generally quicker due to proximity to local emergency services.

Overall Summary of Risk Assessment

Overall, the route is considered moderately safe for heavy vehicles carrying hazardous materials, with peak risks during monsoonal rains and typical peak traffic periods. Road quality is adequate but caution is advised in rural stretches and during adverse weather. Thorough planning, adhering to regulations, and real-time communication are essential for safe transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48363, 77.88780	30 KM/Hr
7	Turn	Medium	11.48446, 77.88787	30 KM/Hr
8	Turn	Medium	11.49451, 77.88489	30 KM/Hr
9	Turn	High	11.55614, 77.89764	15 KM/Hr
10	Turn	High	11.58318, 77.90659	15 KM/Hr

Emergency Locations